



Modernizing Telecom Service Orchestration with Data Engineering

Table of Contents

- Introduction to Telecom Service Orchestration
- Telecom Service Orchestration using Data Engineering Tools
- Summary & Q&A

Presented By
Ganashree M D
DEFT-JULY-2026

Introduction: What is Telecom Service Orchestration?

The Core Objective:

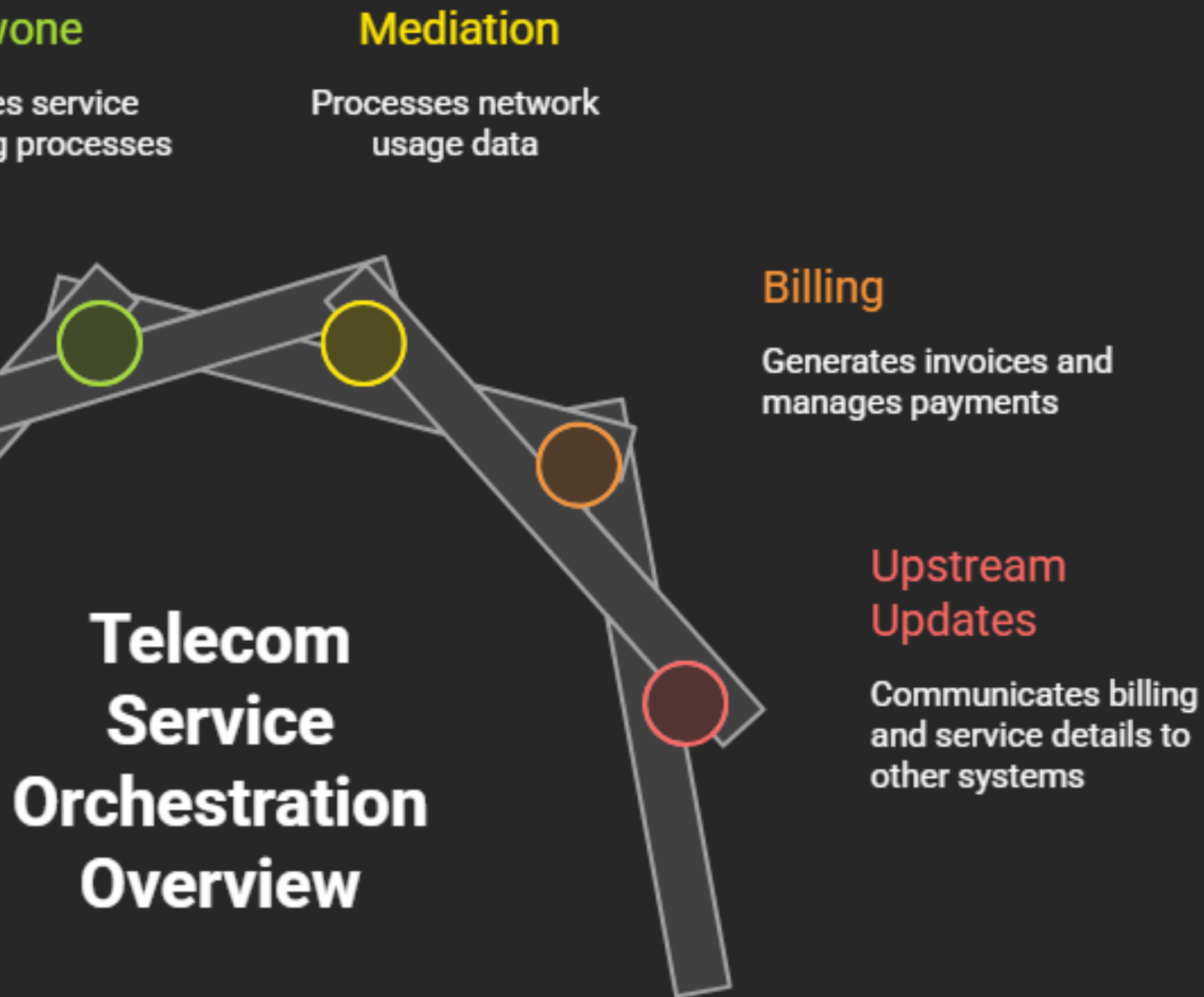
- Automated bridge connecting business operations (**BSS**) with network execution (**OSS**).

The Dual-Layer Ecosystem:

- **BSS (Business Support)**: Manages CRM orders, customer journeys, and billing monetization.
- **OSS (Operations Support)**: Handles technical network execution and **FlowOne** provisioning

The End-to-End Orchestration Flow:

- **Order Capture**: Customer places an order in BSS / CRM.
- **Fulfillment**: Request triggers **FlowOne** to provision network resources.
- **Mediation**: Network generates raw binary usage logs (CDRs) that are decoded.
- **Monetization**: Usage is billed and synced back to BSS to complete the loop.



Telecom Service Orchestration using Data Engineering Tools

Order Capture (BSS CRM)

- **Sqoop** pulls customer records & **Kafka** streams new orders instantly upon plan changes.

Network Activation (FlowOne OSS)

- **Airflow** triggers FlowOne provisioning and tracks job execution., while **Databricks** logs network topology state.

Usage Collection (Mediation)

- **NiFi** grabs raw network usage files on the fly & streams them straight into the **Data Lake (AWS S3 / HDFS)**.

Binary Processing

- **Spark** reads raw binary from the Data Lake, decodes it in parallel, and saves structured **Parquet** files back to storage.

Billing & Loopback

- Decoded data is checked by **Impala** in the lake, loaded into **Snowflake**, transformed via **dbt**(like aggregations,matching records,checking duplicates), and synced back to **CRM**.

Automating the Telecom Pipeline with Data Engineering



Catching
Customer
Orders

Use Apache Sqoop to pull customer data and Kafka to instantly stream new orders.



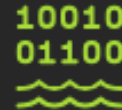
Automating
Network
Activation

Apache Airflow triggers FlowOne to activate services while Databricks tracks live network status.



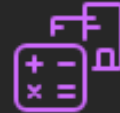
Collecting
Usage Data

Apache NiFi grabs raw usage files and Kafka streams them in real-time.



Translating
Technical Data

Kafka and NiFi automatically decode binary files and match them to customer profiles.



Calculating
Final Bill

Snowflake stores data, dbt calculates charges, Impala runs checks, and the bill syncs to the CRM.

Thank You

Summary: Bridging BSS, FlowOne OSS, and Mediation with modern Data Engineering tools (Kafka, Nifi, sqoop, spark, PySpark, Snowflake, dbt, Airflow).

Open Floor: Questions, ideas, or feedback on applying this architecture to telecom pipelines.